Appendix IV
Layer Analysis, Ablation, and Propagation Sensitivity

Supplementary Material for the IEEE Internet of Things Journal Submission

1 Purpose of This Appendix

This appendix explains why the paper uses four relation layers and how each layer contributes to the final model. It expands the layer-density analysis, aggregation-support figure, ablation table, and propagation-depth sensitivity.

2 Layer Meaning

The four relation layers are designed to separate different mechanisms of RPL behavior:

- **Routing layer**: parent-child DODAG structure. This is the layer most directly affected by DRA because the attack changes parent-selection incentives.
- **Link-quality layer**: weighted version of the observed parent-child support. It distinguishes low-cost and high-cost parent relationships.
- **Temporal layer**: similarity in delay, packet-delivery behavior, and route dynamics across observation windows.
- **Trust/anomaly layer**: similarity in rank inconsistency, parent-metric deviation, and delivery loss.

3 Layer Structure

Table 1: Mean structural properties of the four RPL relation layers.

Layer	Mean edges	Density	Edge share
Routing	49.00	0.040	0.023
Link quality	49.00	0.040	0.023
Temporal	885.78	0.723	0.410
Trust/anomaly	1154.90	0.943	0.544

The routing and link-quality layers are sparse because each node has a limited parent-child structure. The temporal and trust/anomaly layers are denser because they connect nodes by similarity. If all edges are collapsed into one graph, the dense layers dominate the message-passing support and the model loses the distinction between physical routing relations and behavioral similarity relations.

4 Aggregation-Support Figure

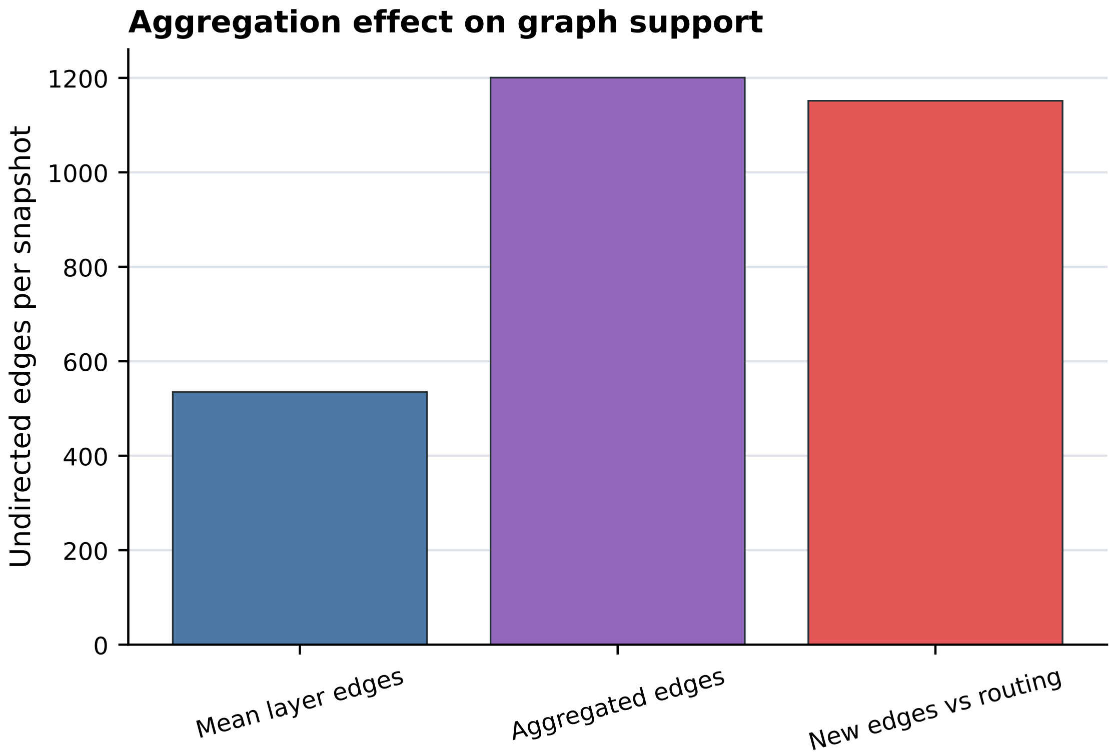

Figure 1: Aggregation-support analysis. Collapsing relation layers increases graph support and hides relation identity.

The figure explains why Agg-GCN is not equivalent to ML-GCN. Agg-GCN receives one combined support, so a message can come from a routing neighbor, a link-quality neighbor, or an anomaly-similar node without preserving which relation produced that edge. ML-GCN keeps the four channels separate and only fuses them after relation-specific convolution.

5 Layer Ablation

Table 2: Single-layer and leave-one-layer-out ML-GCN ablation.

Variant	BAcc	F1	FPR
All layers (ML-GCN)	**0.721**	**0.531**	0.183
Routing only	0.719	0.509	0.265
Link-quality only	0.706	0.495	0.262
Without temporal	0.693	0.481	0.258
Without routing	0.677	0.469	0.212
Without link quality	0.671	0.471	**0.150**
Temporal only	0.525	0.337	0.801
Trust/anomaly only	0.517	0.341	0.967

The ablation indicates that routing and link-quality information are highly informative for DRA detection, which is expected because decreased-rank attacks manipulate routing attractiveness. However, the full multilayer model obtains the best F1-score, showing that temporal and anomaly information add useful context when combined with the routing layers. The temporal-only and anomaly-only variants perform poorly because behavioral similarity alone is too broad and can connect benign nodes with similar noisy dynamics.

6 Propagation Sensitivity

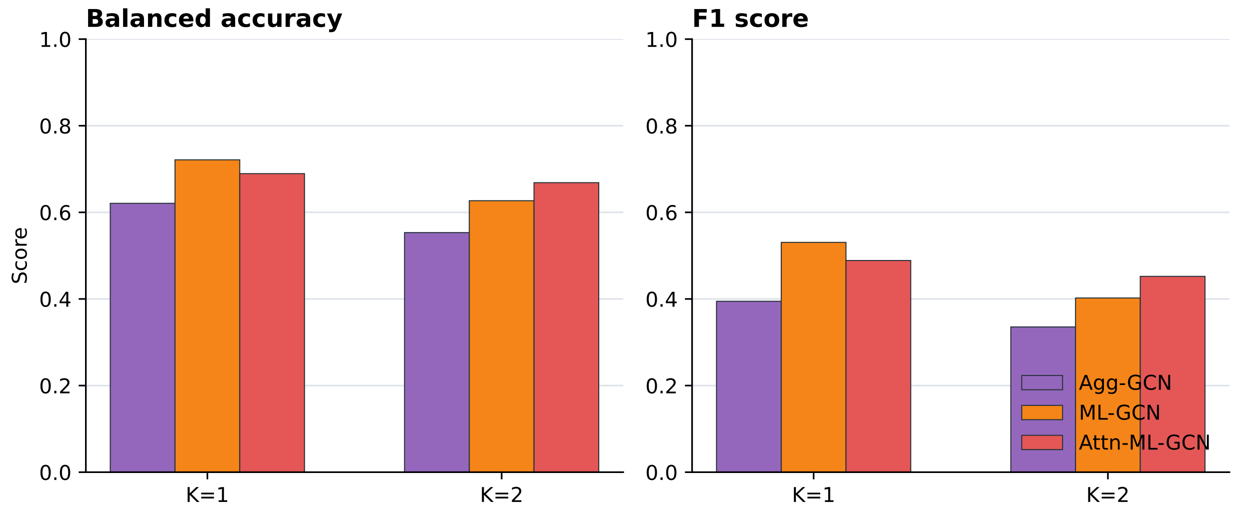

Figure 2: Propagation-depth sensitivity for graph-based models.

The sensitivity experiment compares one-hop and two-hop propagation. In this benchmark, two-hop propagation reduces the detection quality of the graph models. The reason is practical: the LLN snapshots are small, and DRA evidence can be localized around parent-child and nearby behavioral relations. Additional propagation mixes information from wider neighborhoods and can blur the difference between malicious and benign nodes. This supports the paper's choice of one-hop relation-specific message passing.